

MATA37H3 — Week 02

The Fundamental Theorem of Calculus

3 hours • OpenStax, *Calculus Volume 2*, §1.3 (The Fundamental Theorem of Calculus); paired with MAT137 (Calculus with Proofs) Unit 8 lecture-slide prompts on antiderivatives, functions defined by integrals, and FTC Parts 1 & 2 — MAT137 is the University of Toronto’s official exclusion-equivalent of MATA37H3

Recap and orientation

Unit 7 defined $\int_a^b f(x) dx$ as a limit of Riemann sums — a definition that is rigorous but nearly impossible to compute directly for anything but the simplest functions. This unit introduces the **Fundamental Theorem of Calculus (FTC)**, which converts definite-integral computation into an *antiderivative-lookup* problem: no more limits of sums needed. FTC has two parts — Part 1 says differentiating an integral gets you back the original function; Part 2 says you can evaluate a definite integral by plugging endpoints into any antiderivative. Together they say, informally, “integration and differentiation undo each other.”

Why this matters

Every integral you compute from Unit 9 onward — by substitution, by parts, with partial fractions — is really just “find an antiderivative, then invoke FTC Part 2.” Misunderstanding FTC is also one of the most common sources of subtle errors on rigorous exams: confusing “an antiderivative” with “the antiderivative $\int_0^x f$ ”, forgetting the chain rule when the upper limit is itself a function of x , and misapplying FTC to functions that are only piecewise continuous.

DEFINITION Antiderivative and Indefinite Integral (MAT137 Unit 8)

A function F is an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for every $x \in I$. If F is one antiderivative of f on I , then every antiderivative of f on I has the form $F(x) + C$ for some constant C — this is why we write $\int f(x) dx = F(x) + C$ (the **indefinite integral**), with the understanding that C ranges over all real numbers.

Watch for: the “ $+C$ ” fact requires I to be a *single interval*. On a domain with a gap (like $\mathbb{R} \setminus \{0\}$), two antiderivatives can differ by a *different* constant on each piece — this is exactly the trap in Worked Example 1 below.

KEY FACT Functions Defined by an Integral (MAT137 Unit 8)

An expression like $F(x) = \int_a^x f(t) dt$ (with x as the *upper limit*, and a dummy variable t inside) defines a genuine function of x : for each fixed x , the integral is a single number. This is a completely valid way to define a function, **even when f has no elementary antiderivative** (e.g. $F(x) = \int_0^x e^{-t^2} dt$). It is *not* valid to have the same free variable appear both as the bound of integration and as the integration variable, e.g. $\int_0^x \frac{x}{1+t^8} dt$ mixes them; the variable

of integration must be a dummy (like t), never the same letter as the limit.

KEY FACT Fundamental Theorem of Calculus, Part 1 (OpenStax §1.3, Theorem 1.4)

If f is continuous on $[a, b]$, and $F(x) = \int_a^x f(t) dt$, then

$$F'(x) = f(x) \quad \text{for every } x \text{ in } [a, b].$$

In words: the derivative of “the integral of f from a fixed point to a variable point x ” is just $f(x)$ itself. This is the precise sense in which integration and differentiation are inverse operations.

KEY FACT FTC Part 1 with a Function in the Limit (Chain Rule Extension)

If $F(x) = \int_a^{g(x)} f(t) dt$ for a differentiable function g , then by FTC Part 1 combined with the chain rule,

$$F'(x) = f(g(x)) \cdot g'(x).$$

If the **lower** limit is the variable instead, $\int_{g(x)}^b f(t) dt = -\int_b^{g(x)} f(t) dt$, so differentiating picks up an extra minus sign: $\frac{d}{dx} \int_{g(x)}^b f(t) dt = -f(g(x)) g'(x)$. If **both** limits depend on x , split the integral at any convenient constant point and differentiate each piece separately.

KEY FACT Fundamental Theorem of Calculus, Part 2 — the Evaluation Theorem (OpenStax §1.3, Theorem 1.5)

If f is continuous on $[a, b]$ and F is **any** antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a),$$

often written $F(x) \Big|_a^b$. This is what makes definite integrals computable: find *any one* antiderivative, evaluate it at the two endpoints, subtract. It does not matter which antiderivative you pick — the arbitrary constant C always cancels in the subtraction.

KEY FACT General Solution to $G'(x) = f(x)$, $G(x_0) = y_0$ (MAT137 Unit 8)

If f is continuous on $\mathbb{R}$ and we want the *unique* function G with $G'(x) = f(x)$ **and** a prescribed value $G(x_0) = y_0$, the answer is

$$G(x) = y_0 + \int_{x_0}^x f(t) dt.$$

This single formula answers “which antiderivative?” — a bare antiderivative of f is only ever determined up to $+C$; pinning down one specific value $G(x_0) = y_0$ removes that ambiguity and produces exactly this formula, not $\int_0^x f(t) dt$ unless $x_0 = 0$ and $y_0 = 0$.

DEFINITION Mean Value Theorem for Integrals & Average Value (OpenStax §1.3, Theorem 1.3)

If f is continuous on $[a, b]$, there exists at least one point $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

The right-hand side, $\frac{1}{b-a} \int_a^b f(x) dx$, is called the **average value** of f on $[a, b]$ — this theorem says a continuous function always *attains* its own average value somewhere on the interval.

Worked Examples

EXAMPLE 1

The most misunderstood antiderivative

Find $\int \frac{1}{x} dx$, and explain why $F_1(x) = \ln x$ alone is not a correct general answer.

$F_1(x) = \ln x$ has domain $(0, \infty)$ only, so it cannot be the antiderivative on all of $\mathbb{R} \setminus \{0\}$. Instead consider $F_3(x) = \ln|x|$: for $x > 0$, $F_3(x) = \ln x$ so $F'_3(x) = \frac{1}{x}$; for $x < 0$, $F_3(x) = \ln(-x)$ so by the chain rule $F'_3(x) = \frac{1}{-x} \cdot (-1) = \frac{1}{x}$. So $F'_3(x) = \frac{1}{x}$ on **both** pieces of the domain, giving

$$\int \frac{1}{x} dx = \ln|x| + C.$$

But since the domain $\mathbb{R} \setminus \{0\}$ is *not* a single interval, the constant is allowed to be **different on each side of 0**: the most general antiderivative is actually

$$G(x) = \begin{cases} \ln|x| + C_1 & x > 0 \\ \ln|x| + C_2 & x < 0 \end{cases}$$

for independently chosen C_1, C_2 . This is why $F_4(x) = \ln|2x| = \ln|x| + \ln 2$ is *also* a valid antiderivative on $x > 0$ — it just corresponds to $C_1 = \ln 2$ — and does not contradict the “ $+C$ ” rule, because that rule only guarantees a single shared constant on a connected interval. ■

EXAMPLE 2

FTC Part 1, straightforward and with the chain rule

Find $F'(x)$ for (a) $F(x) = \int_2^x \sin(t^3) dt$, and (b) $G(x) = \int_1^{x^2} \frac{\sin t}{t^2} dt$.

(a) By FTC Part 1 directly (the upper limit is bare x), $F'(x) = \sin(x^3)$.

(b) Here the upper limit is $g(x) = x^2$, not x itself, so we need the chain-rule extension: with

$$f(t) = \frac{\sin t}{t^2},$$

$$G'(x) = f(g(x)) \cdot g'(x) = \frac{\sin(x^2)}{(x^2)^2} \cdot 2x = \frac{2x \sin(x^2)}{x^4} = \frac{2 \sin(x^2)}{x^3}. \quad \blacksquare$$

EXAMPLE 3

FTC Part 1 with the variable in the lower limit and both limits

Find $F'(x)$ for $F(x) = \int_{2x}^{x^2} \frac{1}{1+t^3} dt$.

Split at a convenient constant, say $t = 0$:

$$F(x) = \int_{2x}^0 \frac{dt}{1+t^3} + \int_0^{x^2} \frac{dt}{1+t^3} = -\int_0^{2x} \frac{dt}{1+t^3} + \int_0^{x^2} \frac{dt}{1+t^3}.$$

Differentiate each piece with the chain rule (using $f(t) = \frac{1}{1+t^3}$):

$$F'(x) = -f(2x) \cdot 2 + f(x^2) \cdot 2x = \frac{-2}{1+8x^3} + \frac{2x}{1+x^6}. \quad \blacksquare$$

EXAMPLE 4

Reading FTC Part 1 off a piecewise-linear graph

Let f be the piecewise-linear function with $f(0) = 0$, rising with slope 1 to $f(1) = 1$, constant at 1 on $[1, 2]$, falling with slope -1 to $f(3) = 0$ and continuing to $f(4) = -1$, then constant at -1 on $[4, 5]$. Let $F(x) = \int_0^x f(t) dt$. Sketch F and describe F' .

Since F is an antiderivative of f by FTC Part 1, $F' = f$ — so wherever f is *positive*, F is increasing; wherever f is *negative*, F is decreasing; and F has a local max/min exactly where f crosses zero. Computing F by accumulating signed area under f : $F(0) = 0$; on $[0, 1]$, area of a triangle $= \frac{1}{2}$, so $F(1) = \frac{1}{2}$; on $[1, 2]$, add a 1×1 rectangle, so $F(2) = \frac{3}{2}$; on $[2, 3]$, f falls linearly from 1 to 0, adding another triangle of area $\frac{1}{2}$, so $F(3) = 2$ (this is F 's **maximum**, since f changes from $+$ to $-$ there); on $[3, 4]$, f is negative (a triangle from 0 down to -1), subtracting $\frac{1}{2}$, so $F(4) = \frac{3}{2}$; on $[4, 5]$, $f \equiv -1$, subtracting 1 more, so $F(5) = \frac{1}{2}$. So F rises from 0 to a peak of 2 at $x = 3$, then falls to $\frac{1}{2}$ at $x = 5$ — and $F'(x) = f(x)$ throughout, confirming FTC Part 1 graphically. $\blacksquare$

EXAMPLE 5

FTC Part 2 — basic evaluation

Evaluate $\int_1^3 (2x^2 - 4) dx$.

An antiderivative of $2x^2 - 4$ is $F(x) = \frac{2x^3}{3} - 4x$ (any choice of C works; take $C = 0$). By FTC Part 2,

$$\int_1^3 (2x^2 - 4) dx = F(3) - F(1) = \left(\frac{2(27)}{3} - 12\right) - \left(\frac{2}{3} - 4\right) = (18 - 12) - \left(\frac{2}{3} - 4\right) = 6 + \frac{10}{3} = \frac{28}{3}. \quad \blacksquare$$

EXAMPLE 6

Area between a function and the x -axis, watching for sign changes

Find the area of the region bounded by $y = \sin x$ and the x -axis on $[0, 2\pi]$.

$\sin x \geq 0$ on $[0, \pi]$ and $\sin x \leq 0$ on $[\pi, 2\pi]$, so **naively** computing $\int_0^{2\pi} \sin x dx$ would let the negative part cancel the positive part — giving the *net signed area*, not the true geometric area. Instead, split and take absolute values:

$$\text{Area} = \int_0^\pi \sin x dx + \left| \int_\pi^{2\pi} \sin x dx \right| = [-\cos x]_0^\pi + \left| [-\cos x]_\pi^{2\pi} \right| = (1 - (-1)) + |-1 - 1| = 2 + 2 = 4. \quad \blacksquare$$

(Direct check: $\int_0^{2\pi} \sin x \, dx = -\cos(2\pi) + \cos 0 = -1 + 1 = 0$ — the signed total, which is *not* the area — confirming why the split was necessary.)

EXAMPLE 7

Uniqueness of the solution to an IVP-style antiderivative problem

Find the unique function H with $H'(x) = e^{\sin x}$ for all $x \in \mathbb{R}$ and $H(1) = -2$.

By the Key Fact on Page 1, with $x_0 = 1$, $y_0 = -2$, $f(t) = e^{\sin t}$:

$$H(x) = -2 + \int_1^x e^{\sin t} \, dt.$$

This is the *only* function satisfying both conditions: any other antiderivative of $e^{\sin x}$ differs from $\int_1^x e^{\sin t} \, dt$ by a constant C , and demanding $H(1) = -2$ forces $C = -2$ exactly (since $\int_1^1 e^{\sin t} \, dt = 0$). Writing $H(x) = \int_0^x e^{\sin t} \, dt - 2$ instead would be **wrong** unless the extra constant happens to work out — check: at $x = 1$ that expression gives $\int_0^1 e^{\sin t} \, dt - 2 \neq -2$ in general, so that guess fails the initial condition. ■

Practice Problems

Read the hint and the caution before you start each problem — *not after*.

Antiderivatives & Indefinite Integrals

1. Compute $\int (3x^8 - 18x^5 + 1) \, dx$ and $\int \sqrt[3]{x} \, dx$ by “guess and check.”

HINT Use the power rule in reverse on each term, and rewrite $\sqrt[3]{x}$ as $x^{1/3}$ before integrating.

CAUTION — before you solve When the exponent is $1/3$, the new exponent after integrating is $4/3$, and you divide by $4/3$ (i.e. multiply by $3/4$) — don’t mishandle the fraction.

Your working:

2. Compute $\int \sec x \tan x \, dx$ and $\int \frac{1}{x+5} \, dx$.

HINT Recall the two separate ‘guess and check’ antiderivative facts for $\sec x \tan x$ and for $\frac{1}{x+5}$.

CAUTION — before you solve Don’t confuse the antiderivative of $\sec x \tan x$ with that of $\tan x$ — they are different functions.

Your working:

3. Find the domain and derivative of $F(x) = \ln |3x|$, and confirm it is a valid antiderivative of $\frac{1}{x}$ on $x > 0$ by identifying the corresponding constant C in $\ln |x| + C$.

HINT Differentiate $\ln |3x|$ with the chain rule to find $F'(x)$, then compare the result to $\ln x + C$ using a log property.

CAUTION — before you solve Don’t forget the chain-rule factor from differentiating the inside function $3x$ — it cancels with the 3 underneath.

Your working:

Functions Defined by Integrals & FTC Part 1

4. Which of these are *valid* ways to define a function $F(x)$ (i.e. the dummy variable and

the free variable are used correctly)? (a) $F(x) = \int_0^x \frac{t}{1+t^8} dt$; (b) $F(x) = \int_0^x \frac{x}{1+t^8} dt$;
 (c) $F(x) = \int_3^0 \frac{t}{1+x^2+t^8} dt$.

HINT For each option, check whether the free variable x appears anywhere inside the integrand as a dummy-variable stand-in.

CAUTION — before you solve The outer variable x appearing somewhere in the integrand doesn't automatically make a definition invalid — only using it *as the variable of integration* does.

Your working:

5. Find $F'(x)$ for $F(x) = \int_5^x \cos(t^2) dt$.

HINT The upper limit here is bare x , so FTC Part 1 applies with no extra chain-rule factor.

CAUTION — before you solve Don't add an unnecessary chain-rule multiplier when the limit is exactly x , not a function of x .

Your working:

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6. Find $F'(x)$ for $F(x) = \int_x^7 \sin^3(\sqrt{t}) \, dt$. (Careful with the variable in the *lower* limit.)

HINT Flip the limits first (introducing a minus sign) so the variable is in the upper position, then apply FTC Part 1.

CAUTION — before you solve Don't forget the sign flip that comes from having the variable in the *lower* limit.

Your working:

7. Find $F'(x)$ for $F(x) = \int_{\sin x}^{e^x} \frac{t}{1+t^8} \, dt$.

HINT Both limits depend on x here, so differentiate the upper-limit and lower-limit contributions separately.

CAUTION — before you solve Don't only differentiate with respect to the upper limit — the lower limit needs its own chain-rule term with a minus sign.

Your working:

8. Let G be an antiderivative of a continuous function f with domain $\mathbb{R}$. Decide true or false, with a brief justification: (a) $G(x) = \int_0^x f(t) dt$ must hold. (b) There exists $C \in \mathbb{R}$ such that $G(x) = C + \int_0^x f(t) dt$.

HINT For (a), think of a specific antiderivative that doesn't equal the bare integral from 0. For (b), this is exactly the '+C' fact restated.

CAUTION — before you solve Don't assume every antiderivative must be expressible with lower limit exactly 0 — the constant can absorb any base point.

Your working:

FTC Part 2 & Applications

9. Evaluate $\int_0^1 (e^x + e^{-x} - \cos(2x)) dx$.

HINT Antidifferentiate each term separately, being careful with the sign on e^{-x} .

CAUTION — before you solve The antiderivative of e^{-x} is $-e^{-x}$, not e^{-x} — watch the chain-rule sign.

Your working:

10. Evaluate $\int_{\pi/4}^{\pi/3} \sec^2 x \, dx$.

HINT Recall which trig function is the antiderivative of $\sec^2 x$.

CAUTION — before you solve Don't confuse the antiderivative of $\sec^2 x$ (which is $\tan x$) with the derivative of $\sec x$ (which is $\sec x \tan x$).

Your working:

11. Find the area of the bounded region enclosed between $y = x^2 + 3$ and $y = 3x + 1$.

HINT Find where the two curves intersect first, then check which curve is on top on that interval before setting up the integral.

CAUTION — before you solve Don't integrate top-minus-bottom in the wrong order — check which function is larger on the interval, or you'll get a negative area.

Your working:

12. Find the average value of $f(x) = 9 - x^2$ on $[0, 3]$, and find a value $c \in [0, 3]$ where $f(c)$ equals that average value (Mean Value Theorem for Integrals).

HINT Compute the average value with the formula $\frac{1}{b-a} \int_a^b f(x) dx$, then solve $f(c) = \text{that average}$ for c .

CAUTION — before you solve Don't forget to divide by $b - a$ — reporting the bare integral as the average value is a common shortcut error.

Your working:
